

Case Swarm

Architecture and Data Flows

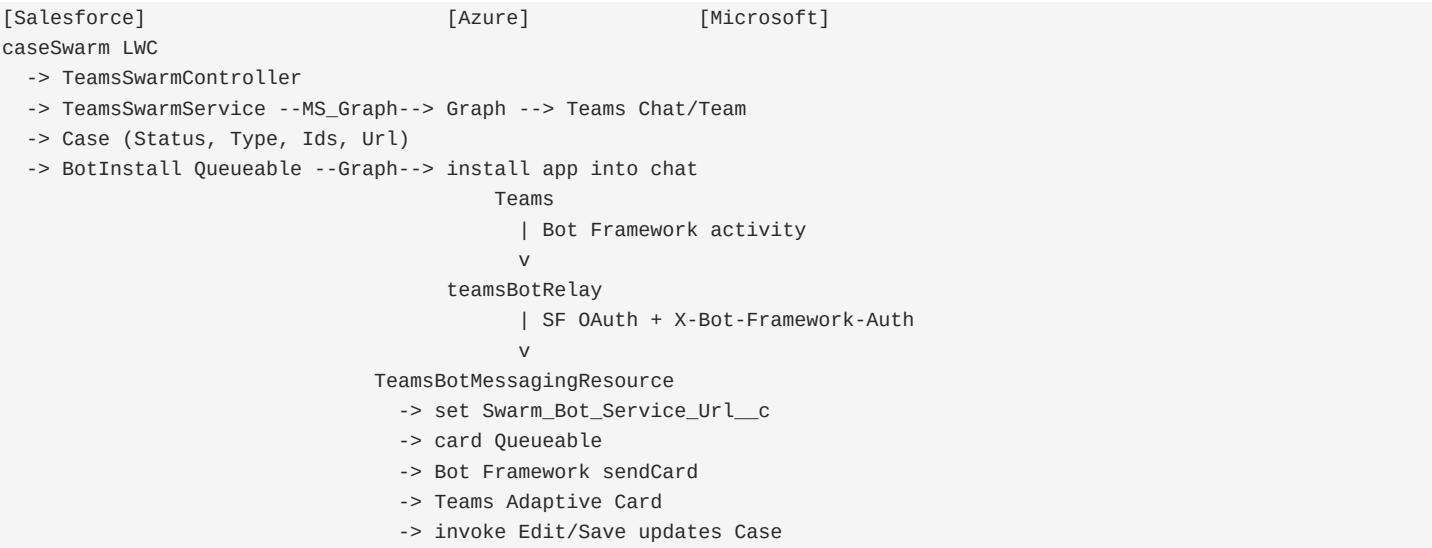
Generated 2026-07-23

Interactive Mermaid diagrams: docs/Case-Swarm-Architecture-and-Data-Flows.md. This PDF is a printable companion. Covers (1) Case swarming alone and (2) Case swarming with Salesforce Apex chat polling and Swarm_Message__c creation. Azure live bridge is out of scope here (see Case-Swarm-Chat-Bridge-Options-and-Limits).

1. Case swarming alone

Creates a Team or Chat in Microsoft Teams, updates Case swarm fields, and for Chat swarms installs the bot and posts an Adaptive Card. No Swarm_Message__c transcript.

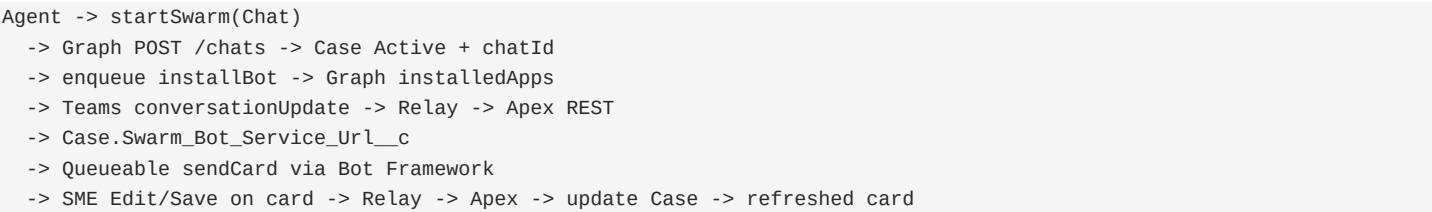
Architecture



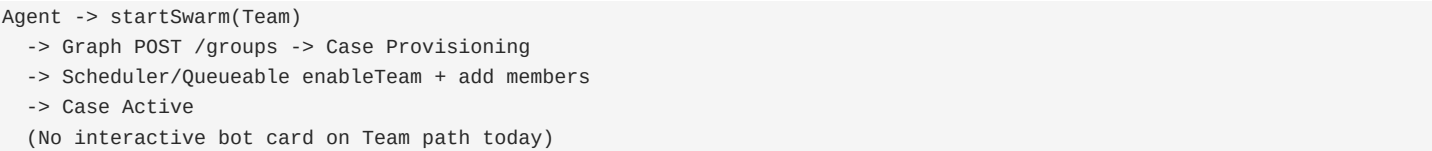
Case fields

Field	Purpose
Swarm_Status__c	Provisioning / Active / Failed
Swarm_Type__c	Team or Chat
Swarm_Team_Id__c	Team/group id or chat id
Swarm_Group_Id__c	M365 group id (Team path)
Swarm_Team_Url__c	Open in Teams link
Swarm_Bot_Service_Url__c	Bot serviceUrl (Chat path)
Swarm_Bot_Install_Error__c	Non-fatal bot install error

Chat swarm sequence



Team swarm sequence



2. Case swarming + SF polling chat sync

When Use_Azure_Chat_Bridge__c = false (or Azure session falls back). Same swarm provision as section 1, plus live chat UI that polls Apex every ~5s. Swarm_Message__c is written DURING the session.

Architecture

Swarm provision (section 1) unchanged.

Live chat (SF poll mode):

```
caseSwarmChatCore
|-- every ~5s --> TeamsSwarmChatController.syncMessages
|
|               -> Graph GET /chats/{id}/messages
|               -> upsert Swarm_Message__c (Inbound; skip bot rows)
|
|-- send -----> TeamsSwarmChatController.sendMessage
|               -> Bot Framework sendText
|               -> insert Swarm_Message__c Outbound (immediate)
|               -> Teams shows message as bot
```

Send from Salesforce

```
LWC sendMessage(caseId, body)
-> Bot Framework POST activity "AgentName: body"
-> Teams
-> INSERT Swarm_Message__c Outbound (Teams_Message_Id = bot:activityId)
-> return stored list to LWC
```

Cost: 1 Apex callout + 1 DML per send

Receive from Teams (poll)

```
SME posts in Teams
... wait until next LWC poll (~5s) ...
LWC syncMessages(caseId)
-> Graph list messages
-> upsert human messages as Inbound
-> skip application/bot Graph rows
-> LWC renders; unread badge if new Inbound
```

Cost: 1 Apex Graph callout per poll tick (+ DML when new)

NOTE: No Graph webhook - Salesforce PULLS on a timer.

Exact code locations - SF chat poll

Runs only when useAzureBridge is false (CMDT Use_Azure_Chat_Bridge__c = false, or Azure session fallback). No Graph webhook is involved.

Step	Location	What
1 Interval	caseSwarmChatCore.js APEX_SYNC_INTERVAL_MS=5000 (~L33-34)	5s cadence
2 Timer	startPolling() (~L349-360)	setInterval -> syncQuietly()
3 Branch	syncQuietly() (~L413-419)	Apex path: fetchApexSync()
4 LWC call	fetchApexSync() (~L433-435)	syncMessages({ caseId })
5 Apex entry	TeamsSwarmChatController.syncMessages (~L199-202)	upsertFromGraph(false)
6 Callout	TeamsSwarmService.listChatMessages (~L253-260)	GET .../chats/{id}/messages
7 Persist	upsertFromGraph in ChatController	Upsert Swarm_Message__c Inbound

CALL CHAIN (SF poll only):

```
caseSwarmChatCore.startPolling()           // setInterval every 5000 ms
-> syncQuietly()                           // useAzureBridge ? Azure : Apex
-> fetchApexSync()                         // SF poll path only
-> TeamsSwarmChatController.syncMessages(caseId)
-> upsertFromGraph(caseId, false)
```

```
-> TeamsSwarmService.listChatMessages(chatId, 50)
-> Http.send GET callout:MS_Graph/.../chats/{id}/messages
-> upsert Swarm_Message__c
```

Azure bridge ON: same timer, but syncQuietly uses fetchAzureHistory()
(browser -> Azure -> Graph). Does NOT call syncMessages each tick.

What creates Swarm_Message__c

Event	Writer	Timing
Agent sends from LWC	sendMessage insert	Immediate
SME posts in Teams	syncMessages upsert	Next poll (~5s)
Bot/application Graph rows	Skipped in syncMessages	N/A
Adaptive Card edit	Case fields only	No chat row

3. Side-by-side

Concern	Swarm alone	Swarm + SF chat poll
Creates Team/Chat	Yes	Yes
Bot + Case card	Chat yes	Chat yes
Live SF-Teams messaging UI	No	Yes
Swarm_Message during session	No	Yes
Graph list from Apex while open	No	Every ~5s
Bot send from Apex	Card only	Card + chat text

4. Flag reminder

Use_Azure_Chat_Bridge__c	Chat live path
false	Section 2 - SF Apex poll + immediate Swarm_Message__c
true	Azure chatHistory/chatSend; transcript only via persistTranscript on close

5. Related code

Area	Path
Start swarm	TeamsSwarmController, TeamsSwarmService, caseSwarm
Bot / card	TeamsBotMessagingResource, install/card Queueables
Relay	relay/src/functions/teamsBotRelay.js
SF chat poll (exact)	See section Exact code locations - caseSwarmChatCore L349-435 -> syncMessages -> listChatMessages L253-260
Transcript object	Swarm_Message__c

6. Webhook?

No Microsoft Graph change-notification webhook is used for chat sync. Inbound messages are discovered by polling (Apex syncMessages or Azure chatHistory). Bot Framework does POST activities to the Azure relay for bot install and Adaptive Card invoke - that is the bot messaging endpoint, not a Graph chat-message webhook.